

Ihtishamul Haque

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Education

Toronto Metropolitan University

September 2020 — April 2025

Bachelor of Science — Computer Science

Toronto, Ontario

- Relevant Coursework: Data Structures & Algorithms, Databases, Operating Systems, Artificial Intelligence, Computer Networks

Relevant Experience

Product Development Engineering Intern

May 2023 — August 2024

Advanced Micro Devices

Markham, Ontario

- Analyzed product data using **JMP** to identify failing patterns and optimize test limits, driving informed quality improvements.
- Streamlined IP testing in product characterization with **Python** and **SQL**, automating data extraction, visualization, and reporting.
- Automated wafer unit estimation, eliminating manual calculations and **cutting calculation time by over 25%**.
- Organized and led **Python** tutorials for engineers, allowing for teams to independently operate internal tools.

Freelance Graphic Designer & Website Developer

December 2016 — June 2020

Self-employed

Toronto, Ontario

- Designed and developed personal portfolio websites using **React** and **HTML/CSS**, tailored to each client's specifications.
- Created promotional graphics for various companies using Photoshop, aligning designs with brand identity and marketing goals.
- Maintained consistent communication with clients, setting clear deadlines, sharing progress, and applying feedback.

Projects

Chessmate — Functional chess engine written in **C++**, uses a **Python** GUI to display real-time gameplay. Engine and game logic were developed entirely from scratch. Uses a **functional programming** approach in core game logic to reduce side effects and complexity. Improved the engine performance by implementing alpha-beta pruning in the search algorithm.

PyBoard — Full-stack calendar application developed entirely in **Python**. Created a custom user interface with interactive calendar functionalities. Integrated **Firebase** for cloud-based data storage, allowing for cross-device data synchronization. Deployed as a standalone app for daily desktop use.

Courtside AI — Computer vision model built to recognize basketball movements, leverages the YOLOv5 model to perform **real-time object detection**. Preprocessed large volumes of basketball video data using **OpenCV** for frame extraction, significantly improving training efficiency. Deployed user interface using **Flask**.

Xv6+ OS — Extended the Xv6 OS using **C** to support additional features and optimizations. Implemented a subset of POSIX-compliant shell coreutils. Added priority-based scheduler to improve CPU utilization. Optimized memory usage by implementing demand paging and page replacement mechanisms.

Technical Skills

Programming Languages: Python, C, C++, TypeScript, Lua, Rust, Haskell, Elixir

Libraries & Frameworks: PyTorch, TensorFlow, OpenCV, Pandas, React, Node.js, Flask, Django

Tools & Platforms: Git, Linux, Visual Studio, Docker, Google Cloud Platform, MongoDB, JMP, PowerBI, Jira, Confluence